ELEC3104: Digital Signal Processing Lab

Dr Phu Le & Professor Eliathamby Ambikairajah

School of Electrical Engineering and Telecommunications

UNSW, Sydney, Australia

Term 3, 2021

Topic 3 – Pole Zero and Filter Response

Exercise 3.1: Consider the following filter:

$$H(z) = \frac{z^8 - 0.9z^7 - 1.375z^6 - 0.975z^5 + 4.8125z^4 - 0.975z^3 - 1.375z^2 - 0.9z^1 + 1}{z^8}$$

Using MATLAB,

a. Find the poles and zeros of this filter and plot them on the z-plane
b. Verify that the zeros of this filter are reciprocal to each other.
c. Plot the phase response of this filter to identify its special property
d. Find the impulse response of this filter
e. The input of this filter is x[n]=[1,3,5,7,9,11,9,7,5,3,1] (n=0,1,2,…10). Find the output of the filter

Topic 3 – Magnitude and Phase Response

Exercise 3.2: a digital filter is given by the following transfer funtuion

$$H(z) = \frac{1-2z+3z^2}{3-2z+z^2}$$

Using MATLAB,

a. Plot the magnitude response of the system.
b. Find the zeros and poles of this system, plot the poles and zeros on the z-plane.
c. Comment on the magnitude response and the pole-zero locations of this filter.

Topic 3 – Phase and Group Delay

Exercise 3.3: A digital filter has eight poles locating at the origin and eight zeros locating at

Zeros	$0.75 \pm j\,0.25$; $-0.5 \pm j\,0.5$; $1.2 \pm j\,0.4$; $-1 \pm j$

Using MATLAB,

a. Plot the pole-zero on z-plane and a frequency response (magnitude and phase) of this filter. Comment on its phase response and the location of zeros.
b. Calculate the group delay of this filter

Topic 3 – Min/Max Phase Filters

Exercise 3.4: The digital filters A, B, and C have four poles locating at the origin and four zeros locating at

	Zeros Location
Filter A	$0.75 \pm j\,0.25$; $-0.5 \pm j\,0.5$
Filter B	$1.2 \pm j\,0.4$; $-0.5 \pm j\,0.5$
Filter C	$1.2 \pm j\,0.4$; $-1 \pm j$

Using MATLAB,

a. Plot the pole-zero on the z-plane and the frequency response (magnitude and phase respones) of each filter.
b. Are there any differences in their magnitude responses?
c. Comment on the phase lag of each of these filters.
d. Classify each of the above filters as either maximum phase, mixed phase or minimum phase filters

Topic 3 – Magnitude and Phase Response

Exercise 3.5: A digital filter is given by the following transfer function

$$H(z) = \frac{0.2}{1-0.8z^{-1}}$$

1. Find the cut-off frequency and bandwidth of the filter
2. Using MATLAB,
    a. Compute and plot the magnitude response of the system.
    b. Write code segment to automatically compute the cut-off frequency and bandwidth of the filter. Compare with the result in question 1

6